

OceanEmbed

Three-dimensional ocean temperature from satellites alone

Smart India Hackathon 2026

INCOIS Problem Statement 01

Progress report

06 September 2026

Domain	5-30N, 45-105E (0.25 deg, daily)
Depth levels	15 standard levels, 0-1000 m
Training record	777 days 2022-01-01 to 2024-10-31
Inputs	SST, SSS, SLA, surface currents, surface wind -- satellite observable only
Training label	GLORYS12V1 reanalysis
Independent check	ARGO floats (never used in training)

Headline result -- measured against real ARGO floats

1.24 C

RMSE vs ARGO floats

1.16 C

GLORYS vs same floats

+7%

Gap to the reanalysis

0.886

Correlation vs GLORYS

ARGO floats never entered training. Held-out days fall chronologically after every training day.

Where the project stands

what has been demonstrated, and what has not

The problem

Satellites see only the ocean surface. Everything that matters for cyclone intensification, fisheries and marine heatwaves happens below it -- and the subsurface is measured by roughly 4,000 ARGO floats for the entire world ocean, which in our basin means one profile every few hundred kilometres every ten days. The operational alternative, a physics assimilation such as GLORYS, needs a supercomputer and runs with a delay.

What we built

A learned mapping from a 9x9 patch of surface fields to the full temperature profile at 15 standard depths, at 0.25 degrees, daily. It trains against GLORYS, runs in seconds on one GPU, and is checked against ARGO floats that never entered training.

What the numbers say

- Against real ARGO floats -- instruments that physically descended through the water column and never entered training -- our RMSE is 1.24 C. GLORYS, the supercomputer reanalysis we are distilled from, scores 1.16 C against those same floats. We are 7% worse than it, from satellites alone.
- Mean correlation 0.886 and RMSE 0.88 C across all 15 depths on held-out days.
- +16% better than climatology on average -- climatology being the per-location mean profile, which needs no satellite data at all and is the honest thing to beat.
- Thermocline (75-300 m) RMSE 1.38 C. This band is where skill above climatology is largest and where surface-only methods have nothing to say.

What we are careful not to claim

- We do not beat GLORYS. GLORYS is our teacher; the most we can be is a fast, cheap, satellite-only approximation of it.
- Below about 500 m the surface carries almost no information. The model correctly falls back to climatology there rather than inventing signal.
- The record is still partial -- November and December are missing from every year, so no complete annual cycle has been seen yet.
- The alerting layer is a proof of concept. Nothing is issued or transmitted to anyone.
- We carry a warm bias against ARGO that peaks in the thermocline. It is stated in full on the validation page rather than averaged away.

Progress so far

Every increase in the training record has been measured rather than assumed. The full history is three pages on.

Independent validation against ARGO floats

the only numbers here not measured against our own teacher

Everything else in this report compares us against GLORYS, which trained us. ARGO floats are real instruments that physically descended through the water column. They never entered training. This is therefore the only genuinely independent measurement in the document.

quantity	value
n profiles	40000
n argo days	18
our rmse vs argo degC	1.242
glorys rmse vs argo degC	1.157
excess over glorys degC	0.085
excess over glorys pct	7.313

Against ARGO, by depth

depth m	n	ours	GLORYS	gap	our bias	corr
0	40000	0.88	0.85	+0.02	+0.28	0.699
5	40000	0.88	0.85	+0.02	+0.28	0.699
10	40000	0.93	0.89	+0.04	+0.36	0.711
20	40000	1.13	1.11	+0.02	+0.37	0.560
30	40000	1.07	1.05	+0.02	+0.44	0.669
50	40000	1.26	1.56	-0.31	+0.46	0.649
75	40000	1.70	1.99	-0.29	+0.64	0.545
100	40000	2.35	2.07	+0.29	+1.36	0.494
125	40000	2.41	2.00	+0.42	+1.53	0.545
150	40000	2.17	1.79	+0.38	+1.06	0.605
200	40000	1.53	1.15	+0.39	+0.63	0.734
300	40000	0.75	0.59	+0.16	+0.27	0.898
500	40000	0.55	0.49	+0.06	+0.29	0.913
700	40000	0.51	0.54	-0.03	+0.28	0.932
1000	40000	0.52	0.44	+0.08	+0.31	0.897

Highlighted rows are depths where we beat the reanalysis against the floats. Treat those as gridded-product smoothing rather than a genuine win: the honest claim there is parity, not superiority.

The bias, stated plainly

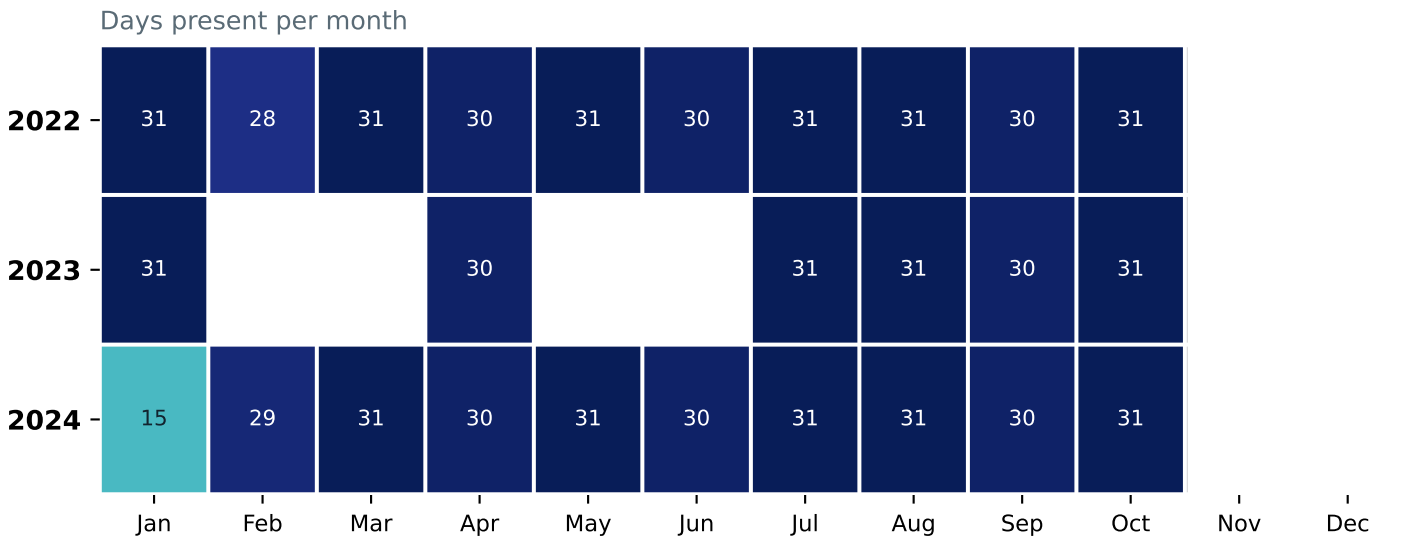
We run systematically WARM against the floats at every level, and the bias peaks at +1.53 C at 125 m -- in the thermocline, which is exactly the band we claim as our strength. A warm bias there means we would overestimate heat content and, downstream, cyclone heat potential. This is a real defect rather than a rounding artifact. It is also correctable: the floats provide an independent reference to fit a per-depth bias correction against, which is queued work, not a limitation of the approach.

GLORYS itself does not match ARGO exactly. The gap between our error against ARGO and GLORYS's own error against ARGO is the honest measure of what our fast approximation costs.

Data coverage

what has actually arrived, by month

777 daily analysis days, 2022-01-01 to 2024-10-31. Each cell is the number of days present for that month. Blank cells are months still being collected.



Collection status

source	assignment	days inmonths still owed
Laptop 1	2022	30411, 12
Laptop 2	2023	18402, 03, 05, 06, 11, 12
Laptop 3	2024	28911, 12

Variables in the harmonized cube

variable	dimensions	missing
sst	time x lat x lon	51.8%
sss	time x lat x lon	52.6%
sla	time x lat x lon	51.7%
ucur	time x lat x lon	54.7%
vcur	time x lat x lon	54.7%
uwind	time x lat x lon	11.6%
vwind	time x lat x lon	11.6%
temp	time x depth x lat x lon	58.2%
land_mask	lat x lon	0.0%

Missing fractions near 50-60 per cent are land plus the deep levels under the continental shelf, not gaps in the satellite record. Land is masked out before any training sample is drawn.

Method

how a surface patch becomes a depth profile

Pipeline

raw products	harmonize	samples	model	products
-----	-----	-----	-----	-----
SST (OSTIA)	common 0.25 deg	9x9 patch of	CNN encoder	D26
SSS (Multiobs)	grid, daily,	7 surface vars	128-d latent	TCHP
SLA (DUACS)	same calendar,	centred on each	depth-attn	MLD
currents	land masked	ocean cell	head	anomaly
wind (ERA5)			15 depths	heatwave flag
GLORYS -----	label only -----	> profile at 15 z		

Why a patch and not a point

Thermocline depth responds to mesoscale eddies, and an eddy is a spatial structure -- a sea-level bump tens of kilometres across with a rotating current around it. A single pixel cannot express that. The 9x9 window lets the encoder see the gradient and the curl, which is what actually indicates the isotherms have been pushed down.

Why the model predicts an anomaly, not a temperature

Predicting absolute temperature means most of what the model has to learn is the climatology itself -- the fact that 1000 m here is always about 7.6 C. That is a lookup table, not inference, and any wobble the model added at depth made it worse than simply quoting the table. Measured: -161 per cent skill against climatology at depth.

So the target is temperature minus the local mean profile, and the climatology is added back at the end. Predicting zero now IS climatology, which makes zero the floor instead of something to fall below. Weight decay pulls the answer toward zero exactly where the surface carries no information, which is the truthful behaviour at depth. The climatology is built from training days only, so the holdout cannot leak into it.

Splits

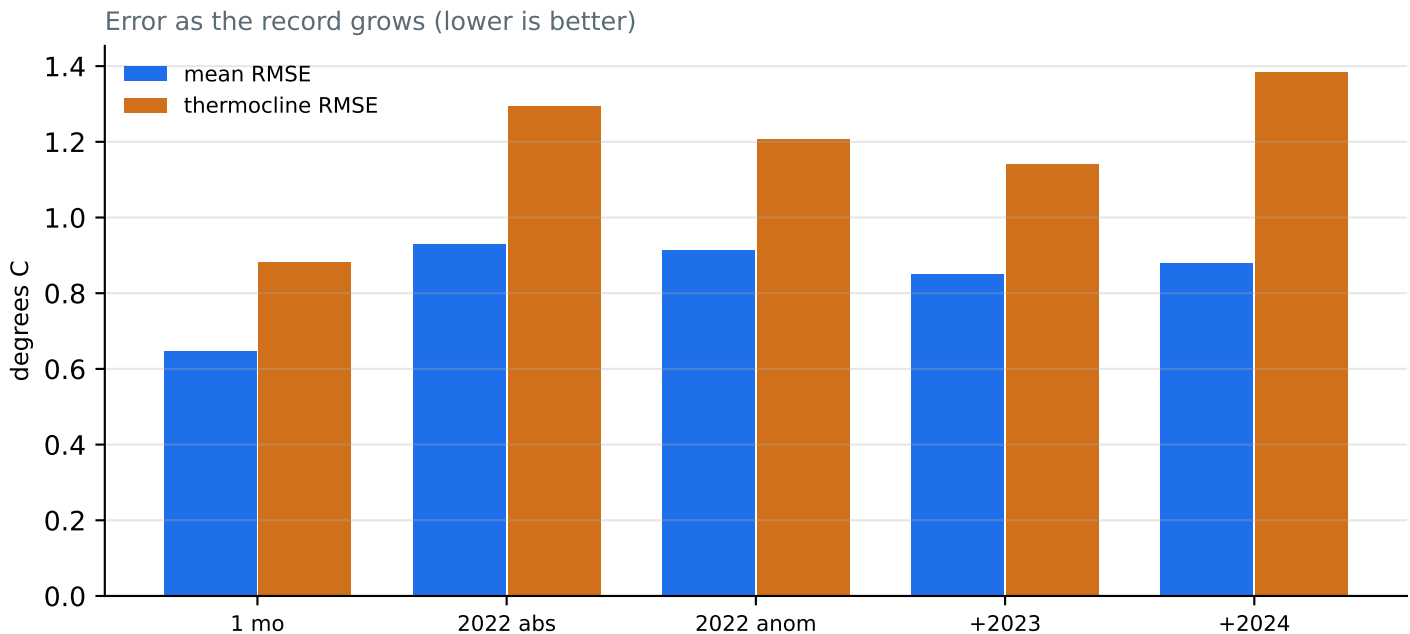
Chronological. Every held-out day comes after every training day. A random split would let the model see Tuesday and Thursday and be asked about Wednesday, which is interpolation rather than prediction, and would inflate every number in this report.

```
held-out profiles : 93600
latitude range   : 6.2 to 24.0 N
longitude range  : 50.5 to 96.8 E
```

Progress across training runs

each row is a real run, not a projection

run	train days	corr	RMSE C	vs clim	thermo C
Oct 2024 only (1 month)	22	0.868	0.65	+25%	0.88
2022 Jan-Oct, absolute target	219	0.804	0.93	+5%	1.29
2022 Jan-Oct, anomaly target	219	0.831	0.91	+27%	1.21
2022 + partial 2023	353	0.849	0.85	+22%	1.14
2022 + 2023 + 2024 (current)	561	0.886	0.88	+16%	1.38



Reading this table honestly

- The holdout changes between rows, because it is always the most recent block of a growing record. Absolute errors are comparable across rows; skill percentages are only comparable within the same holdout.
- The single-month run scored well because its holdout was the same season as its training. That is the easiest possible test, and it is no longer quoted as evidence.
- The large jump in skill came from switching the target to anomalies, not from more data. More data then improved the absolute error.

Accuracy against depth

where the model earns its keep, and where it does not

depth m	corr	RMSE C	bias C	vs clim	vs ARGO	GLORYS
0	0.872	0.54	-0.01	+41%	0.88	0.85
5	0.872	0.54	-0.01	+41%	0.88	0.85
10	0.872	0.54	-0.01	+41%	0.93	0.89
20	0.874	0.57	+0.01	+39%	1.13	1.11
30	0.864	0.69	+0.08	+32%	1.07	1.05
50	0.827	1.15	+0.43	+24%	1.26	1.56
75	0.775	1.76	+0.98	+23%	1.70	1.99
100	0.818	1.91	+1.30	+22%	2.35	2.07
125	0.868	1.68	+1.18	+25%	2.41	2.00
150	0.886	1.43	+0.91	+22%	2.17	1.79
200	0.907	1.03	+0.58	-2%	1.53	1.15
300	0.954	0.49	+0.20	+4%	0.75	0.59
500	0.980	0.24	+0.08	+1%	0.55	0.49
700	0.967	0.28	-0.01	-20%	0.51	0.54
1000	0.954	0.30	+0.09	-50%	0.52	0.44

The first four columns are measured against GLORYS on held-out days. The last two are measured against real ARGO floats: our error, and the reanalysis's own error against the same instruments.

Bands

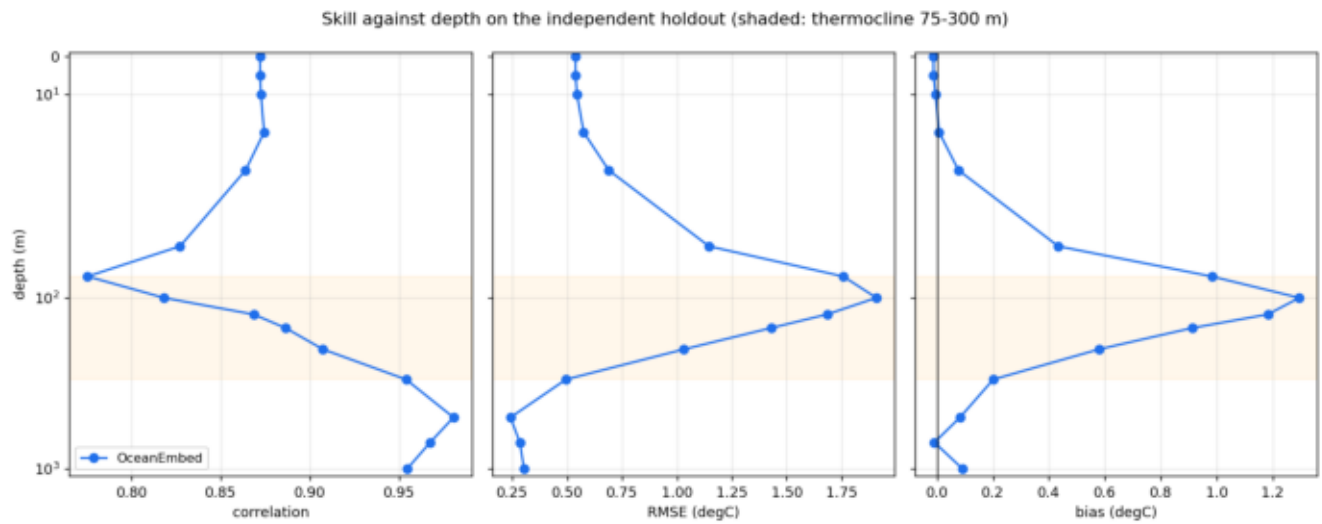
band	corr	RMSE C
mixed layer 0 50m	0.864	0.67
thermocline 75 300m	0.868	1.38
deep 500 1000m	0.967	0.28

What the shape means

- Near the surface the error is small but so is the skill above climatology -- SST is nearly the answer already, so there is little for a model to add.
- In the thermocline the error is largest in absolute terms and the skill above climatology is highest. That is the honest signature of real surface-to-depth inference: the layer is hard, and we are the only thing beating the average there.
- Below 500 m the skill returns to about zero. The surface genuinely does not constrain the deep ocean on daily timescales, and a model claiming otherwise would be fabricating.

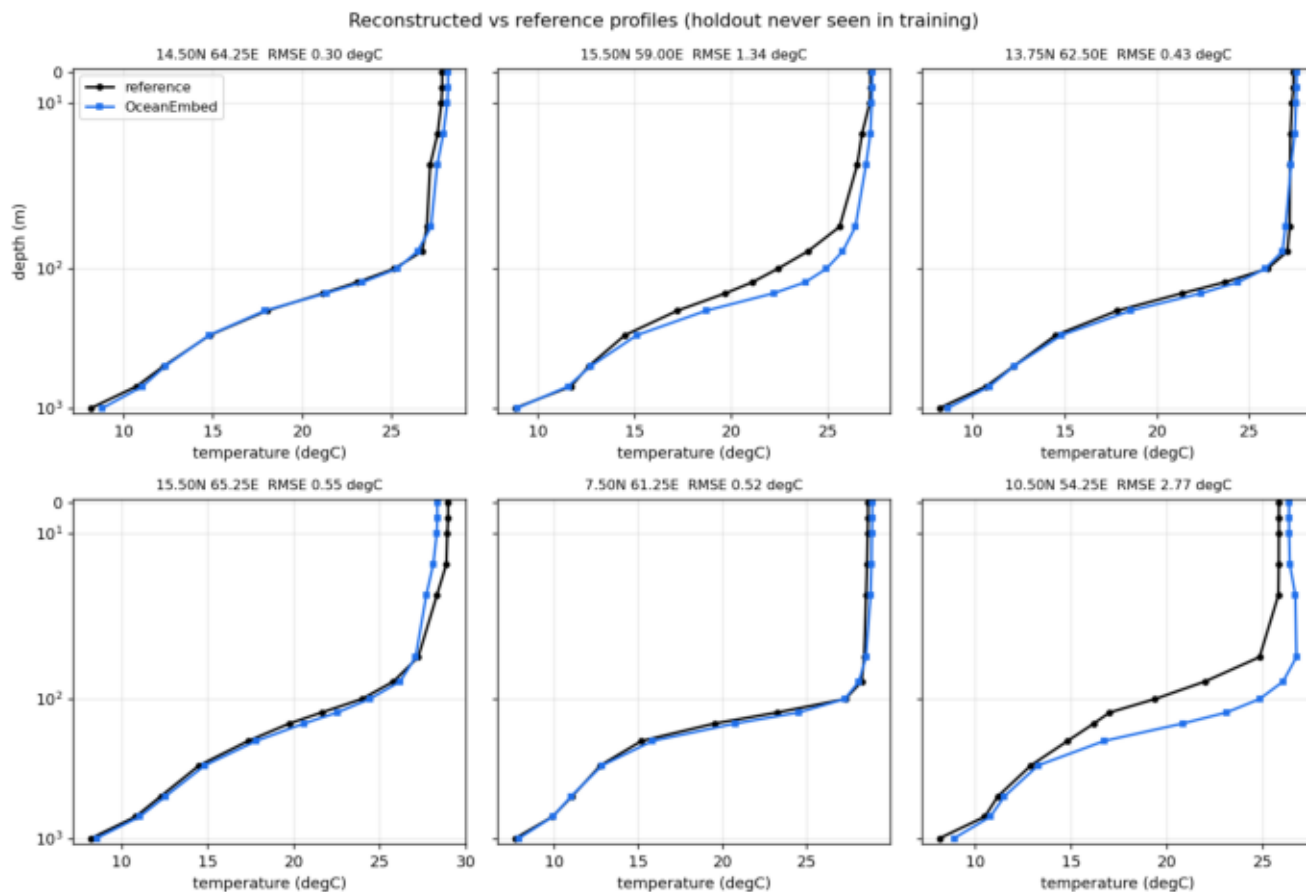
Skill against depth

correlation, RMSE and bias at every level, model against baseline



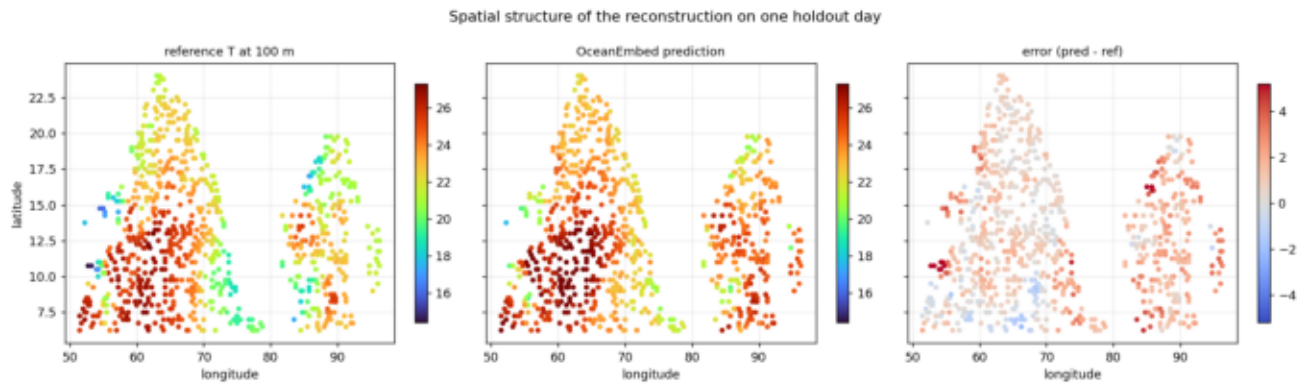
Reconstructed profiles

individual held-out profiles, predicted against GLORYS



Spatial structure of the error

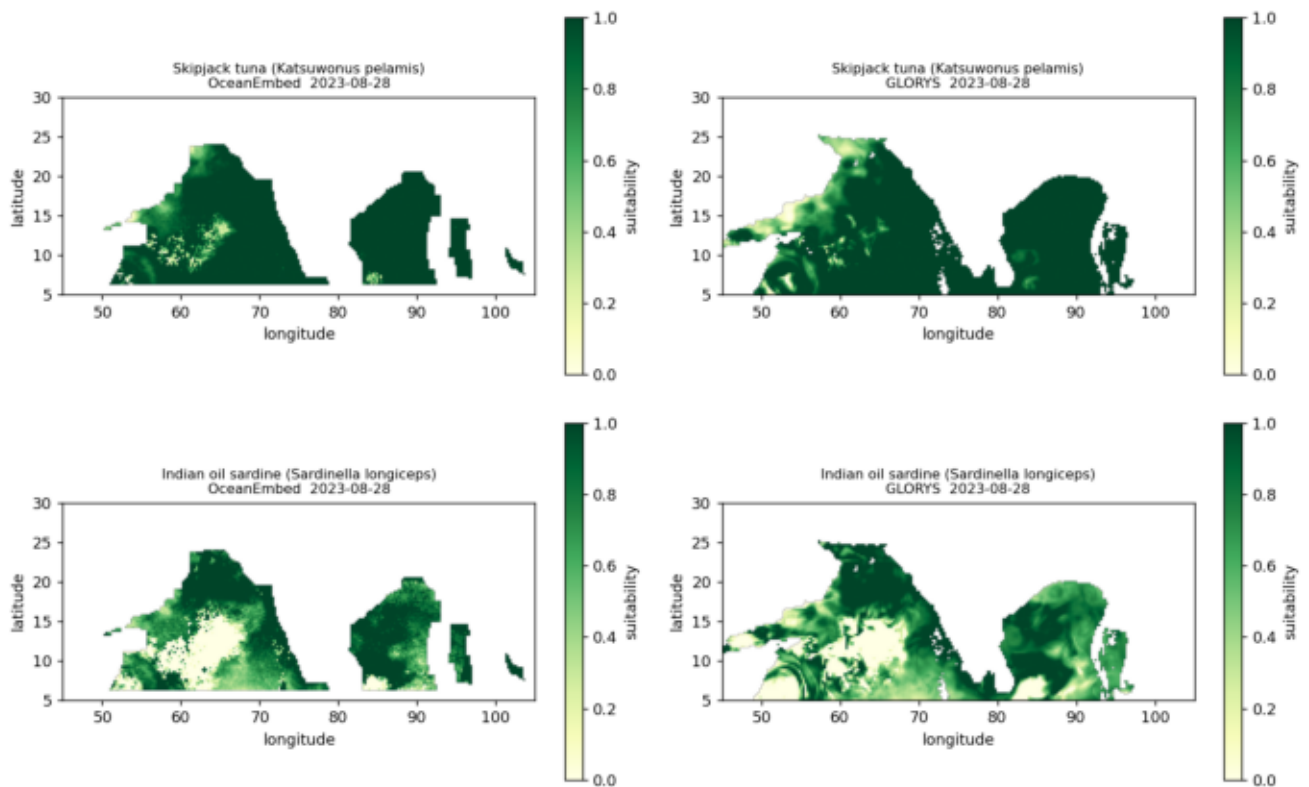
one held-out day at 100 m: reference, prediction, difference



Thermal habitat suitability

predicted against the same score computed from GLORYS

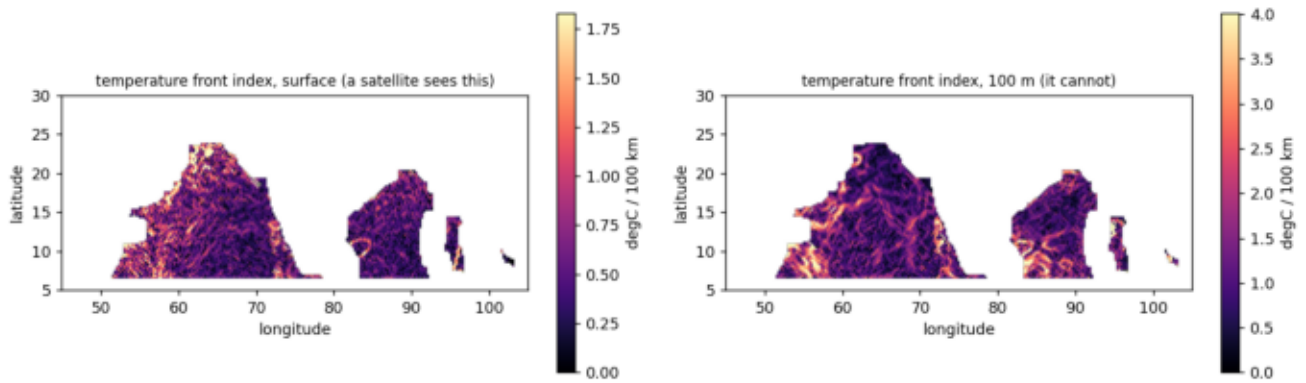
Thermal habitat suitability -- advisory on water properties, not a claim about fish



Thermal fronts, surface and at depth

the subsurface field is the one no satellite can provide

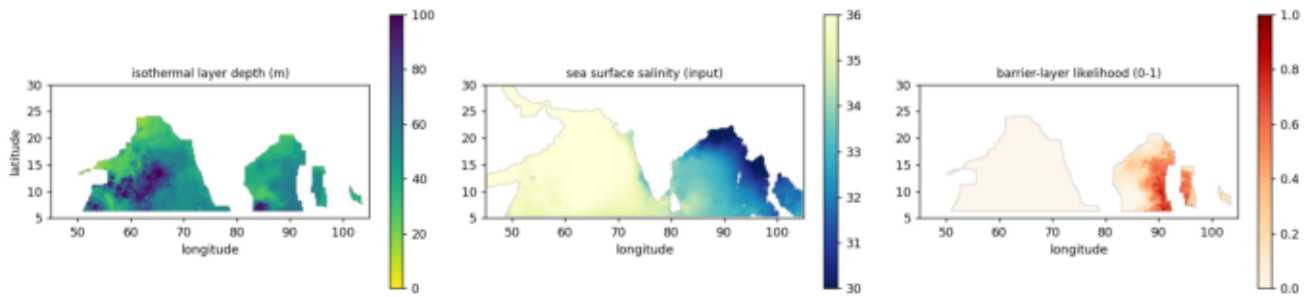
Thermal fronts on 2023-08-28 -- fish aggregate along sharp gradients



Barrier-layer conditions

fresh surface water over a deep isothermal layer

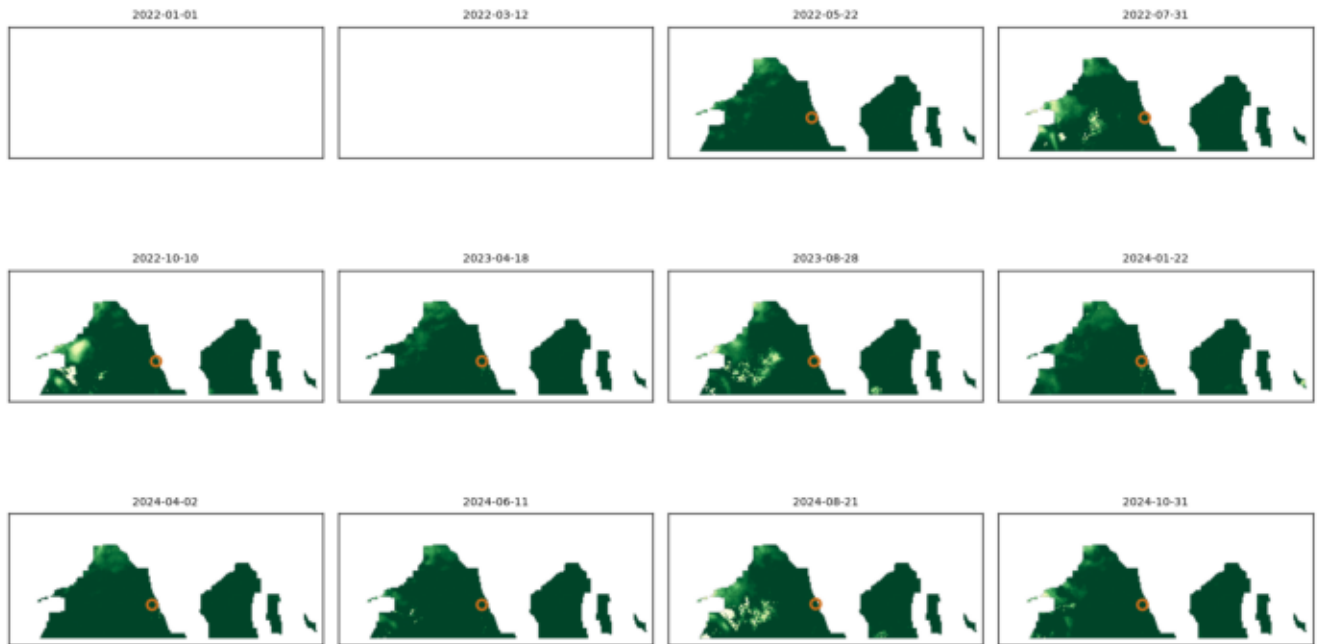
Barrier-layer conditions, 2023-08-28 -- fresh water over a deep isothermal layer (Bay of Bengal signature)



Seasonal shift of the suitable zone

centroid of cells scoring above 0.6, across the record

Seasonal shift of the suitable zone -- Skipjack tuna (*Katsuwonus pelamis*)
circle marks the centroid of cells scoring above 0.6



Operational products

forecasters act on these, not on a temperature field

All three are computed from the predicted profile and compared against the identical computation performed on GLORYS, on the same held-out days. D26 is the depth of the 26 C isotherm; TCHP is the heat stored above it, the quantity used to anticipate rapid cyclone intensification; MLD is the mixed layer depth.

product	unit	n	corr	RMSE	bias
D26	m	93600	0.815	18.13	+9.09
TCHP	kJ/cm2	93600	0.845	18.24	+6.16
MLD	m	93600	0.613	17.43	+1.27

Would we have raised the same alarm?

TCHP above roughly 50 kJ/cm2 is the rule-of-thumb threshold for rapid intensification. Precision and recall at that line matter more than RMSE.

threshold kJ/cm2	precision	recall	F1	n above
50	0.882	0.929	0.905	61434
80	0.579	0.894	0.703	22449
100	0.269	0.603	0.372	6087

Subsurface anomaly and marine heatwave detection

depth m	anom corr	anom RMSE	recall	precision	base rate
0	0.662	0.54	0.355	0.340	2.1%
30	0.580	0.69	0.196	0.274	6.7%
75	0.642	1.76	0.391	0.308	9.8%
100	0.703	1.91	0.600	0.230	5.2%
150	0.717	1.43	0.742	0.283	5.2%
200	0.578	1.03	0.547	0.280	6.2%
300	0.437	0.49	0.481	0.397	8.8%

Detecting a warm anomaly 100 m below the surface, daily, from satellites alone is something surface-only monitoring cannot do at all. A formal marine-heatwave definition also requires five days of persistence, which needs a longer record than we have.

Habitat, fronts and barrier-layer conditions

advisory layers derived from the reconstructed profile

Four layers, each a cheap function of a profile we already predict, and each validated against the identical computation performed on GLORYS. The thermal habitat layer is the same product category as the INCOIS Potential Fishing Zone advisory, which today uses sea surface temperature and chlorophyll and carries no subsurface term at all.

derived layer	unit	n	corr	RMSE	bias
isothermal layer depth	m	65435	0.716	13.606	-1.554
OHC (0-700 m)	kJ/cm2	65435	0.961	92.550	+30.412
TCHP	kJ/cm2	65435	0.918	13.168	-0.833
front index, surface	C/100km	62030	0.263	0.689	+0.259
front index, 100 m	C/100km	60970	0.347	1.217	-0.056
barrier-layer proxy	0-1	65435	0.846	0.109	+0.002
habitat: skipjack_tuna	0-1	65435	0.539	0.132	+0.018
habitat: yellowfin_tuna	0-1	65435	0.675	0.195	+0.035
habitat: indian_oil_sardine	0-1	65435	0.685	0.273	+0.057
habitat: indian_mackerel	0-1	65435	0.598	0.223	+0.042

Agreement on the call a user would actually act on

Not RMSE -- whether we flag the same cells as suitable (score above 0.6) that GLORYS would.

species	precision	recall	F1	base rate
skipjack_tuna	0.974	0.989	0.982	96.3%
yellowfin_tuna	0.847	0.931	0.887	74.0%
indian_oil_sardine	0.788	0.878	0.831	56.2%
indian_mackerel	0.866	0.936	0.900	79.7%

What these layers are, and are not

- Thermal habitat is a transparent threshold-and-taper score on temperature and isothermal layer depth. Nobody trained it on catch data, so it cannot be validated as a catch predictor. It is an advisory about water properties, not a claim that fish are present.
- Fronts are computed at depth as well as at the surface. The surface version is what satellite products already provide; the subsurface version is the one nothing else can give.
- The barrier-layer field is a LIKELIHOOD, not a thickness. A true barrier layer is isothermal layer depth minus DENSITY mixed layer depth, and density needs salinity at depth, which this project does not predict. Reporting metres here would be fabricating a quantity we cannot compute. Adding a salinity head would make it real.
- Species preference bands are broad literature values and should be replaced with regionally tuned ones by a fisheries scientist.

Robustness and ablations

what happens when an input disappears, and what the design is worth

Losing one satellite at a time

Each input channel is blanked in turn and the model re-run. The useful column is not the error -- it is which variable the model actually depends on. A channel whose removal costs nothing was never being used.

missing input	mean RMSE C	change	TCHP RMSE
nothing (full)	0.877	--	18.2
sst	0.934	+6.4%	19.9
sla	0.899	+2.5%	17.1
vwind	0.882	+0.5%	18.5
uwind	0.881	+0.4%	18.4
vcur	0.870	-0.8%	17.8
sss	0.869	-1.0%	16.9
ucur	0.866	-1.4%	18.6

The point is not that accuracy drops -- of course it does. The point is that the system still runs and still returns a calibrated profile. A physics assimilation cannot do that without a full rerun.

Does the spatial patch actually matter?

patch size	mean RMSE C	corr
1x1	0.633	0.872
3x3	0.624	0.860
5x5	0.619	0.879
9x9	0.563	0.885
11x11	0.580	0.888

Best patch: 9. A 1x1 patch is the point-only case -- if it were as good, the whole spatial argument would collapse.

Decision support layer

proof of concept -- nothing here is issued or transmitted

The temperature field is the middle of the system, not the end of it. On top of the reconstruction sit a cyclone-intensification risk index, a coastal exposure calculation over 30 named segments of the Indian coastline, and an alert log.

Risk index inputs

- Tropical cyclone heat potential above 50.0 kJ/cm², the rapid-intensification threshold.
- A requirement that TCHP is also anomalously high for that location. An absolute threshold alone fires almost everywhere in this basin during the warm season -- adding the anomaly condition cut flagged cell-days from about 137,000 to about 21,000.
- Model uncertainty from Monte-Carlo input resampling. A high-heat cell the model is unsure about is downgraded rather than escalated.

Alert log (1283 entries over the evaluated period)

date	coastal segment	level	TCHP
	Ratnagiri	ELEVATED	-
	Goa	ELEVATED	-
	Chennai	HIGH	-
	Mangaluru - Udupi	HIGH	-
	Cuddalore - Puducherry	HIGH	-
	Kozhikode - Kannur	HIGH	-
	Nellore	HIGH	-
	Mumbai - Thane	ELEVATED	-
	Nagapattinam - Delta	HIGH	-
	Port Blair	HIGH	-
	Kochi - Alappuzha	HIGH	-
	Machilipatnam - Krishna	HIGH	-

Decision support only. This shows where the OCEAN could sustain rapid intensification, based on reconstructed subsurface temperature. It does not detect or forecast cyclones, does not account for atmospheric conditions, and is not an evacuation advisory. Proof of concept, not an operational warning product.

What is left

in priority order

Blocking on data

- November and December are missing from every year, so the model has never seen a complete annual cycle. Until then the seasonal coordinate channels stay switched off -- they were measured to hurt when the test season was absent from training.
- Months still outstanding: 2022-11, 2022-12, 2023-02, 2023-03, 2023-05, 2023-06, 2023-11, 2023-12, 2024-11, 2024-12.

Experiments queued

- Temporal context: add 7-day and 30-day mean wind and the 7-day SST change as extra channels. The thermocline responds to accumulated forcing, not to a single day of it. Cheap, and needs no new data.
- Retrospective cyclone replay over a real storm, to show the risk layer responding to an event with a known outcome.
- Re-enable the seasonal channels once complete years are in.

Deliverables

- Web interface: built and deployed -- map, depth and date selection, six layers, 3D seafloor view, alert log.
- This report, regenerated from disk by scripts/09_report.py.
- Slide deck: not started.

Reproducing everything in this document

```
python scripts/01c_merge_harmonized.py "data/incoming/*.nc"
python scripts/02_build_dataset.py --max-per-day 800
python scripts/03_train.py --epochs 150 --weight-decay 1e-3
python scripts/03b_baseline.py
python scripts/04_evaluate.py
python scripts/07_products.py
python scripts/03d_sensor_robustness.py
python scripts/08_risk.py
python scripts/09_report.py
```